

Solution of HW 24

Problem Statement

Solve the least squares system for the given matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad b = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Solution

To find the least squares solution, we solve the normal equation:

$$A^T A x = A^T b.$$

Step 1: Compute $A^T A$ and $A^T b$

$$A^T A = \begin{bmatrix} 35 & 44 \\ 44 & 56 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 76 \\ 100 \end{bmatrix}.$$

Step 2: Write the Augmented Matrix

$$\begin{bmatrix} 35 & 44 & 76 \\ 44 & 56 & 100 \end{bmatrix}.$$

Step 3: Perform Gaussian Elimination

Divide the first row by 35:

$$\begin{bmatrix} 1 & \frac{44}{35} & \frac{76}{35} \\ 44 & 56 & 100 \end{bmatrix}.$$

Final augmented matrix:

$$\begin{bmatrix} 1 & 0 & -6 \\ 0 & 1 & 3 \end{bmatrix}.$$

The least squares solution is:

$$x_1 = -6, \quad x_2 = 3.$$

Problem Statement

Solve the least squares system for the given matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad b = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Solution

The least squares solution is computed as:

$$x = (A^T A)^{-1} A^T b.$$

Step 1: Compute $A^T A$ and $A^T b$

The transpose of A is:

$$A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}.$$

Compute $A^T A$:

$$A^T A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 35 & 44 \\ 44 & 56 \end{bmatrix}.$$

Compute $A^T b$:

$$A^T b = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 76 \\ 100 \end{bmatrix}.$$

Step 2: Compute the Inverse of $A^T A$

To compute $(A^T A)^{-1}$:

$$A^T A = \begin{bmatrix} 35 & 44 \\ 44 & 56 \end{bmatrix}.$$

The determinant is:

$$\det(A^T A) = (35)(56) - (44)(44) = 1960 - 1936 = 24.$$

The inverse is given by:

$$(A^T A)^{-1} = \frac{1}{\det(A^T A)} \begin{bmatrix} 56 & -44 \\ -44 & 35 \end{bmatrix}.$$

Substitute the determinant:

$$(A^T A)^{-1} = \frac{1}{24} \begin{bmatrix} 56 & -44 \\ -44 & 35 \end{bmatrix}.$$

Step 3: Compute $x = (A^T A)^{-1} A^T b$

Multiply $(A^T A)^{-1}$ with $A^T b$:

$$x = \frac{1}{24} \begin{bmatrix} 56 & -44 \\ -44 & 35 \end{bmatrix} \begin{bmatrix} 76 \\ 100 \end{bmatrix}.$$

Perform the multiplication:

$$x = \frac{1}{24} \begin{bmatrix} (56)(76) + (-44)(100) \\ (-44)(76) + (35)(100) \end{bmatrix}.$$

Simplify:

$$x = \frac{1}{24} \begin{bmatrix} 4256 - 4400 \\ -3344 + 3500 \end{bmatrix} = \frac{1}{24} \begin{bmatrix} -144 \\ 156 \end{bmatrix}.$$

Finally:

$$x = \begin{bmatrix} -6 \\ 6.5 \end{bmatrix}.$$

Conclusion

The least squares solution is:

$$x_1 = -6, \quad x_2 = 6.5.$$